A Platform for Spatially Structured Ethological Behavior

Abstract: We propose a modular home-cage platform that converts the cage floor into a programmable foraging environment for freely moving rodents. Through local fabrication, deployment, and support, this resource will enable WashU investigators to perform continuous, spatially structured behavioral experiments compatible with modern neural recording systems.

PROJECT OVERVIEW AND SIGNIFICANCE

Naturalistic behavior provides an essential complement to reductionist preparations for understanding neural circuit function, particularly for linking circuit dynamics to real-world navigation, decision-making, and pathological behavioral states. This perspective has driven growing interest in naturalistic and ethologically grounded paradigms that study behavior in more realistic contexts [1,2]. Such approaches have revealed low-dimensional structure in natural behavior encoded by striatal circuits [3], state-dependent modulation of sensory responses during self-initiated actions [4], and foundational principles of spatial coding in freely moving animals [5]. **In each case, progress depended on new advances in experimental platforms.**

Foraging provides a powerful ethological framework for studying brain–behavior relationships because it naturally integrates spatial navigation, decision-making under uncertainty, and motivation while enabling direct tests of exploration–exploitation tradeoffs and value-based decisions [6–8]. However, natural foraging is difficult to reproduce under controlled laboratory conditions: existing behavioral systems often isolate only fragments of this process, relying on fixed interaction points and discrete trial structures that limit spatially structured behavior [9,10]. As a result, there remains a need for platforms that enable continuous, spatially structured, self-initiated behavior at scale while remaining compatible with neural recordings [11–13].

We address this gap with **a modular home-cage platform that transforms the cage floor into a distributed, programmable foraging environment**, enabling continuous, ethologically grounded behavior in freely moving rodents. The system combines distributed reward delivery and spatial cueing within a scalable, synchronized architecture compatible with chronic neural recordings (Figure 1).

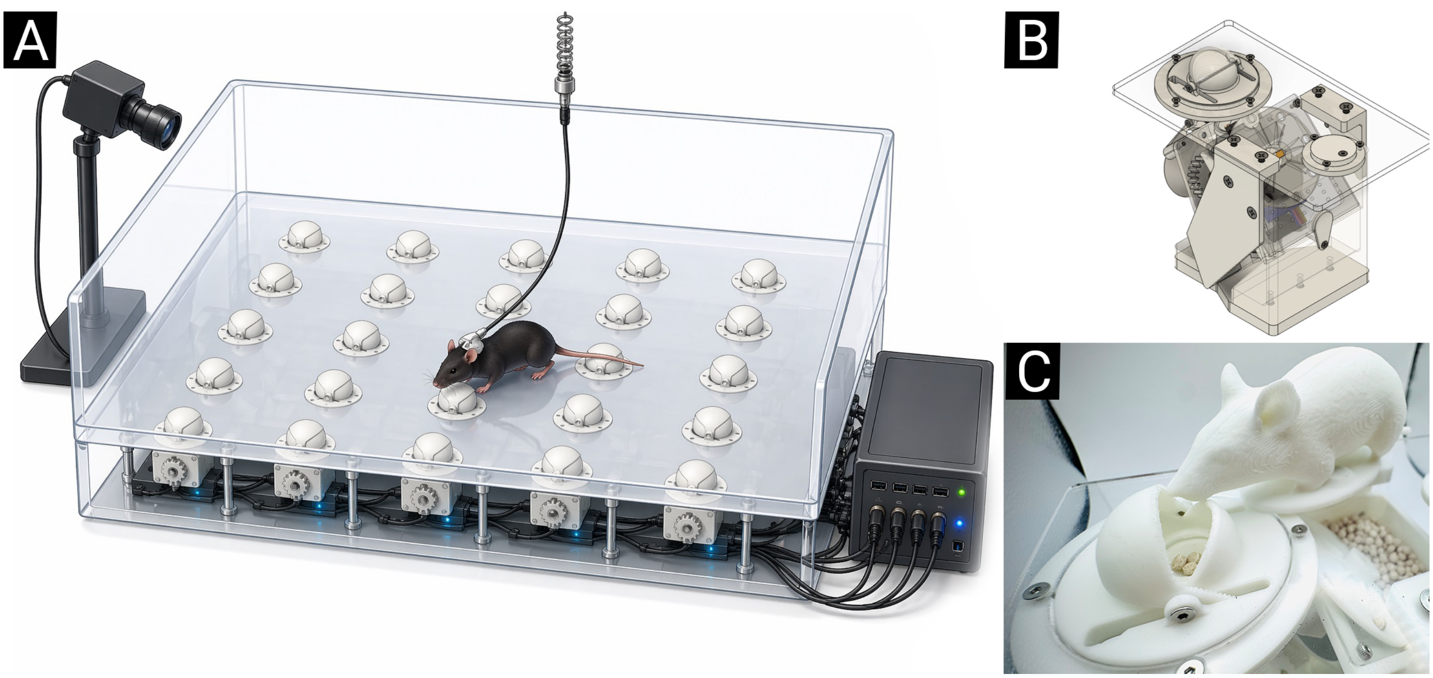

Figure 1: New Resource Concept & Prototype. (A) Arrayed foraging modules conceal and deliver rewards within a spatially distributed environment, enabling free navigation and interaction with hidden, experimenter-controlled reward sites while supporting synchronized neural recordings and video monitoring. (B) 3D design of a single module featuring a gravity-resetting teetering clamshell for covert ground-level pellet delivery with integrated interaction sensing. (C) Working Neurotech Hub prototype showing a mouse interacting with the clamshell reward site.

While foraging provides a natural use case, the platform supports diverse behavioral paradigms through independent control of spatial, motivational, and task variables. By varying module number, spacing, reward schedules, and cueing, the system enables high-throughput, ethologically grounded behavior while bridging a critical gap between naturalistic and tractable neuroscience experiments.

TECHNOLOGY PROPOSAL

We propose a home-cage system in which the floor is composed of an array of independently addressable modules that deliver and conceal food rewards, creating a spatially distributed foraging environment. Each module uses a two-stage pellet delivery mechanism: a single-pellet feeder coupled to a linear actuator that presents reward beneath a low-profile clamshell. Integrated sensors verify discrete reward delivery and capture interactions at each site.

Modules are coordinated over a CAN bus network, a communication standard used in automotive and aerospace systems, enabling scalable and spatially flexible deployment across cage configurations. A central host coordinates experimental control, logs behavioral data, and provides synchronization with external systems.

This architecture creates a **programmable resource landscape**, enabling continuous, self-initiated, and spatially structured behavior with precise experimental control. The platform supports a wide range of questions in systems neuroscience, including spatial navigation and hippocampal coding, decision-making and foraging theory, motivation, learning, and naturalistic behavior.

REQUESTED FUNDS, ANTICIPATED TIMELINE, DISSEMINATION PLAN

The total requested funds for this project are $40,000, distributed across engineering, behavioral benchmarking, and validation efforts required to develop and disseminate the shared resource:

Recipient	Category	Amount
Neurotech Hub	Engineering salary (electronics, firmware, UI, integration)	$15,000
	Fabrication & prototyping equipment	$5,000
Animal Behavior Core	Benchmarking module array (n=9)	$3,000
	Behavioral testing, animals, and technician effort	$7,000
Hengen Lab	Validation module array (n=9)	$3,000
	Electrophysiology validation, animals, and recording support	$7,000
	Total =	**$40,000**

1. Neurotech Hub: Lead engineering, fabrication, and system integration, including development and dissemination of custom electronics and a Linux-based host interface for spatial configuration, cueing, and reward control. Utilize in-house fabrication methods and support adoption through strong documentation with native support for syncing recording systems with ongoing technical oversight and support.
2. Behavior Core: Lead testing against ground-truth paradigms and provide user experience feedback. Work together with Neurotech Hub to develop common task structures (e.g., bandit, reaction time, unrestricted free feed) and deployment schemes to drive broader use and applicability. Provide final analysis report on animal metrics and task engagement.
3. Hengen Lab: Lead in vivo validation using chronic electrophysiology in freely moving mice, including (i) behavioral validation of robust, self-initiated engagement and stable reward interaction dynamics, and (ii) recording validation demonstrating compatibility with long-timescale acquisition and recovery of canonical signals (e.g., hippocampal spatial tuning and striatal/ACC decision-variable encoding) under spatially structured behavior.

The proposed development, validation, and dissemination will occur over a 12-month period. By completion, we will deliver a **fully validated, build-ready behavioral platform** including complete CAD files, bill of materials, and firmware released via GitHub with step-by-step assembly, calibration procedures, and example experimental configurations (e.g., reward schedules, spatial layouts) to enable rapid replication.

The Neurotech Hub will serve as a distribution and support node, providing build services, integration support (e.g., electrophysiology, video), and user onboarding. Multi-lab validation (Hengen and Behavior Core) will establish generalizability and produce reference datasets and analysis examples. Dissemination will include a methods publication, hands-on Neurotech Hub workshops, and ongoing updates through the open-source repository, positioning the platform for immediate local use and scalable external adoption.

REFERENCES

[1] Datta, S. R., Anderson, D. J., Branson, K., Perona, P., & Leifer, A. (2019). Computational neuroethology: A call to action. *Neuron*, 104(1), 11–24.

[2] Krakauer, J. W., Ghazanfar, A. A., Gomez-Marin, A., MacIver, M. A., & Poeppel, D. (2017). Neuroscience needs behavior: Correcting a reductionist bias. *Neuron*, 93(3), 480–490.

[3] Markowitz, J. E., Gillis, W. F., Beron, C. C., Neufeld, S. Q., Robertson, K., Bhagat, N. D., et al. (2018). The striatum organizes 3D behavior via moment-to-moment action selection. *Cell*, 174(1), 44–58.

[4] Niell, C. M., & Stryker, M. P. (2010). Modulation of visual responses by behavioral state in mouse visual cortex. *Neuron*, 65(4), 472–479.

[5] O'Keefe, J., & Dostrovsky, J. (1971). The hippocampus as a spatial map: Preliminary evidence from unit activity in the freely moving rat. *Brain Research*, 34(1), 171–175.

[6] Grima, L. L., Haberkern, H., Mohanta, R., Morimoto, M. M., Rajagopalan, A. E., & Scholey, E. V. (2025). Foraging as an ethological framework for neuroscience. *Trends in Neurosciences*.

[7] Mobbs, D., Trimmer, P. C., Blumstein, D. T., & Dayan, P. (2018). Foraging for foundations in decision neuroscience: Insights from ethology. *Nature Reviews Neuroscience*, 19(7), 419–427.

[8] Kane, G. A., James, M. H., Shenhav, A., Daw, N. D., Cohen, J. D., & Aston-Jones, G. (2022). Rat anterior cingulate cortex continuously signals decision variables in a patch foraging task. *Journal of Neuroscience*, 42(29), 5730–5744.

[9] Matikainen-Ankney, B. A., Earnest, T., Ali, M., Casey, E., Wang, J. G., Sutton, A. K., et al. (2021). An open-source device for measuring food intake and operant behavior in rodent home-cages. *eLife*, 10, e66173.

[10] Choi, J.-S., & Kim, J. J. (2010). Amygdala regulates risk of predation in rats foraging in a dynamic fear environment. *PNAS*, 107(50), 21773–21777.

[11] Murrell, C. L., Legaria, A. A., McCullough, K. B., Nwacha, A., Nasiru, M. O., Ferreira Dias, S. A., et al. (2026). Sex differences in exploration–exploitation strategies during home-cage decision making. *bioRxiv*. https://doi.org/10.64898/2026.04.02.716124

[12] Parks, D. F., Schneider, A. M., Xu, Y., Brunwasser, S. J., Funderburk, S., Thurber, D., et al. (2024). A nonoscillatory, millisecond-scale embedding of brain state provides insight into behavior. *Nature Neuroscience*. https://doi.org/10.1038/s41593-024-01715-2

[13] Xu, Y., Schneider, A., Wessel, R., & Hengen, K. B. (2024). Sleep restores an optimal computational regime in cortical networks. *Nature Neuroscience*. https://doi.org/10.1038/s41593-023-01536-9

[14] McGregor, J. N., Farris, C. A., Ensley, S., Schneider, A., Fosque, L. J., Wang, C., et al. (2024). Failure in a population: Tauopathy disrupts homeostatic set-points in emergent dynamics despite stability in the constituent neurons. *Neuron*. https://doi.org/10.1016/j.neuron.2024.08.00